Energy Conversion MS4092

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Energy Conversion MS4092

Number of problems: 8
Start at: 08.32 PM
Finish at: 08.56 PM
Duration: 24 Minutes

A Diesel engine operates with an ideal cycle that has a compression ratio of 17 and a cutoff ratio of 1.4. If it produces 150 kW, rate of heat addition to the cycle?

- ○ 209.8 kW
- ○ 211.5 kW
- ✗ 229.3 kW
- ○ 212.2 kW

$r = 17 \quad k = 1,4$

$r_c = 1,4$

$\eta = 1 - \frac{1}{17^{1,4-1}} \left[\frac{1,4^{1,4}-1}{1,4(1,4-1)} \right]$

$\eta = 0,654$

$\eta = \frac{W}{Q}$

$Q = \frac{W}{\eta} = \frac{150\ kW}{0,654}$

$Q = 229,36\ kW$

A series configuration of two Carnot engines is designed so that the heat rejected from the first engine is utilized as heat input for the second engine. The temperature limits are 1050 K and 300 K for the hot and cold reservoirs, respectively. If the ratio of the work produced by engine 1 : engine 2 is 1:2, what is the intermediate temperature between the engines? $T_h = 1050\ K \quad T_c = 300\ K$

- ○ 700 K
- ○ 900 K
- ✗ 800 K
- ○ 1000 K

$W_2 = 2W_1$

$T - T_c = 2(T_h - T)$

$T - 300 = 2 \cdot 1050 - 2T$

$3T = 2100 + 300$

$T = 800\ K$

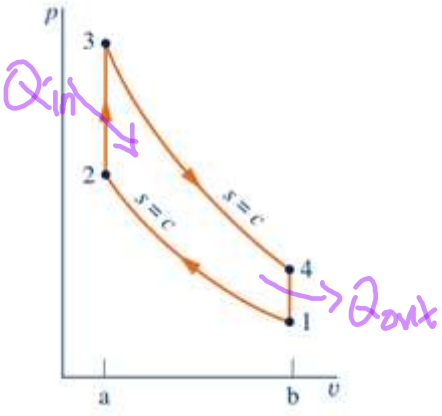
A condenser of AC has surface temperature of 60°C. If the condenser surface is considered as black body, what is the heat flux emitted by the condenser surface? (Stefan-Boltzmann constant = 5.67×10^-8 W.m-2.K-4)

- ○ 545.2 W/m2
- ○ 617.1 W/m2
- ○ 786.2 W/m2
- ⊗ 697.2 W/m2

$60°C + 273 = 333\ K$

$E\sigma T^4 = 1\,(5.67\times10^{-8})(333)^4$

The temperature at the beginning of the compression process of a cold air-standard Otto cycle with a compression ratio of 10 is 27°C, and the maximum temperature of the cycle is 627°C. Determine the rate of heat added per unit mass (kJ/kg), if cv = 0.718 kJ/kg.K



- ○ 151.3 kJ/kg
- ○ 127.5 kJ/kg
- ○ 176.9 kJ/kg
- ⊗ 105.1 kJ/kg

$T_1 = 300\ K = 540\ °R$ $\quad k = 1.4$

$T_3 = 900\ K = 1620\ °R$

$r = 10 = \frac{V_1}{V_2}$

$\frac{V_3}{V_4} = \frac{1}{10}$

$\frac{T_2}{T_1} = \left(\frac{V_1}{V_2}\right)^{k-1} = 10^{0.4} = 2.512$

$\frac{T_4}{T_3} = \left(\frac{V_3}{V_4}\right)^{k-1} = \left(\frac{1}{10}\right)^{0.4} = 0.398$

$T_2 = 2.512\,(T_1) = 2.512\,(300\ K) = 753.6\ K$

$T_3 = 900\ K$

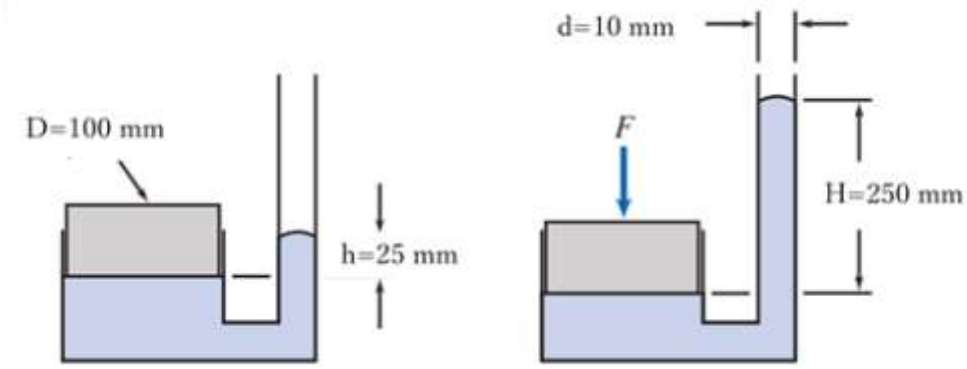
$Q_{in} = C_v \cdot \Delta T = 0.718\ kJ/kg.K\ (900 - 753.6)\ K$

$Q_{in} = 105.11\ kJ/kg$

What will happen to the COP of an air conditioner if the ambient temperature increases to keep the room temperature remain constant?

- ◯ decrease ✓
- ◯ remain constant
- ◯ increase
- ◯ less than 1

The tube shown is filled with water (ρ=1000kg/m^3). Calculate the force applied to the piston. (g = 9.81 m/s^2)



- ◯ 22.07 N
- ⊗ 17.17 N
- ◯ 31.88 N
- ◯ 26.98 N

tekanan yg diteruskan sama

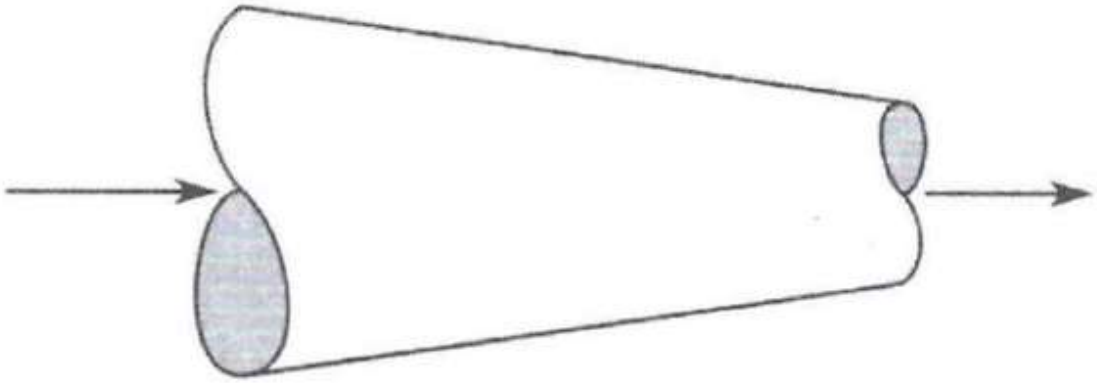
$P = \rho g (H-h) = 1000\,(9.81)\,\frac{(250-25)}{1000}$

$= 2207.25\ Pa$

$F = P\,A = 2207.25\ Pa \times \frac{\pi}{4}(0.1\ m)^2$

$F = 17.33\ N$

A converging nozzle with inlet diameter of 150 mm and outlet diameter of 75 mm shown as follow, has the flowing water at inlet with a velocity of 1.5 m/s and discharged to the atmosphere. With the water is incompressible and friction effect is negligible, what is the gage pressure at the inlet?



- ○ 12.7 kPa
- ○ 10.8 kPa
- ○ 11.3 kPa
- ⊗ 16.8 kPa

$A_1 . V_1 = A_2 . V_2$

$D_1^2 . V_1 = D_2^2 . V_2$

$V_2 = V_1 . \frac{D_1^2}{D_2^2}$

$= 1,5 \left(\frac{150^2}{75^2}\right)$

$V_2 = 6 \text{ m/s}$

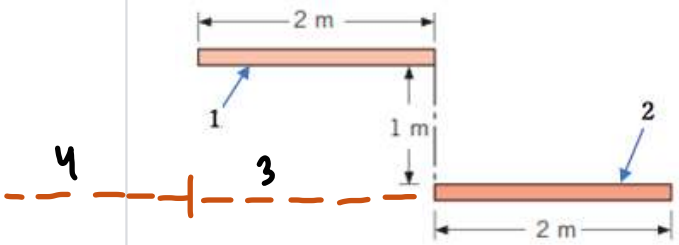
$P_2 = 1 \text{ atm} = 10^5 \text{ Pa}$

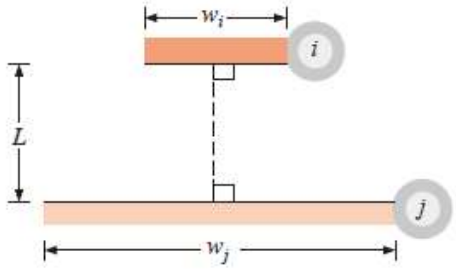
$P_1 + \gamma z_1 + \frac{1}{2} \rho V_1^2 = P_2 + \gamma z_2 + \frac{1}{2} \rho V_2^2$

$P_1 = 10^5 + \frac{1}{2} . (1000) (6^2 - 1,5^2)$

Gage pressure $\rightarrow P_{1g} = P_1 - 10^5$

The parallel planes of infinite extent normal to the page having opposite edges aligned as shown in the following figure. Determine the view factor F12





$$F_{ij} = \frac{[(W_i + W_j)^2 + 4]^{1/2} - [(W_j - W_i)^2 + 4]^{1/2}}{2W_i}$$

$$W_i = w_i/L,\ W_j = w_j/L$$

$W_1 = 2$

$W_3 = 2$

$W_{234} = 6$

- ~~0.163~~
- 0.256
- 0.358
- 0.132

$F_{13} = 0,618$

$F_{1,234} = 0,944$

$F_{12} = \frac{1}{2}(F_{1,234} - F_{13})$

$F_{12} = 0,163$

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